stGradient: Scientific Rationale and Methods

Why, what, and how to read each analysis in the package

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1. Framing: why the tumor boundary?

A solid tumor is not a bag of cancer cells; it is an organ-like structure whose behaviour is shaped by the interface between malignant epithelium and the surrounding stroma. Most of what determines whether an anti-tumor immune response succeeds happens at or near that interface: T cells have to traffic out of vessels in the stroma, cross a fibroblast- and matrix-rich margin, and physically contact tumor cells. Tumors are classically described as *inflamed* (immune cells penetrate the parenchyma), *immune-excluded* (immune cells accumulate in the stroma but fail to enter) or *immune-desert* (few immune cells anywhere) [Chen & Mellman 2017]. The excluded phenotype in particular is a spatial phenotype – it is invisible to bulk or dissociated single-cell sequencing and is associated with resistance to checkpoint blockade [Mariathasan et al. 2018].

Imaging-based spatial transcriptomics (e.g. 10x Xenium) gives us single-cell-resolution expression *with coordinates*, so we can finally ask quantitative questions about this interface:

1. **Where is the boundary?** (ROI selection)
2. **Who is on each side of it?** (cell-type composition inside vs. outside)
3. **How far does the tumor's influence extend into the stroma?** (the three distance metrics and the critical-distance analysis)
4. **How are non-tumor cells organised relative to each other outside the tumor?** (pair correlation function analysis)
5. **Which genes change as a function of distance from the tumor?** (gene–distance analysis)
6. **Are there organised immune niches, such as tertiary lymphoid structures?** (TLS analysis)

stGradient is built around a single organising variable – the **signed distance of every cell to the nearest tumor boundary** (negative inside, positive outside, in microns) – and each analysis below is a different lens on how biology varies along that axis.

A note on statistical units, which applies to every section. In a single tissue section, cells are not independent biological replicates. Neighbouring cells share microenvironment, lineage and segmentation artefacts. P-values computed with cells as the unit of replication (Wilcoxon tests, per-cell regressions) will be extremely small simply because n is large, and should be treated as descriptive rankings rather than inferential statements. Claims about biology should be supported by replication across patients or sections (the patient is the experimental unit) [Squair et al. 2021].

2. ROI selection: defining the tumor boundary

Why do it

Every downstream result in this package is conditional on where the boundary is drawn. "Inside" and "outside" are not properties of the data; they are a model we impose. A boundary that is too loose folds stroma into the tumor and dilutes every inside/outside contrast; a boundary that is too tight mislabels tumor-edge cancer cells as stroma and creates spurious "peritumoral" signal. Getting this step right – and being explicit about how it was done – is the single most important determinant of whether the remaining analyses are interpretable and reproducible.

The scientific question is: *which regions of the section constitute contiguous tumor parenchyma (tumor nests), as opposed to isolated or scattered cancer cells and the stroma between nests?*

What the package does

There are two complementary routes. Both produce an `sf` polygon object that is used by every downstream function.

Manual delineation (`launch_stGradient()`). A Shiny/plotly app plots every cell, coloured by the `is_cancer` annotation, and lets the user draw free-form closed paths, rectangles or circles. Drawn shapes are converted to closed polygons and returned to the R session. This is the right tool when a pathologist's judgement should define the region (e.g. a specific nest, an invasive front, or excluding a necrotic area), or to correct automated boundaries.

Automated delineation (`add_dbscan_polygons()`).

1. Take the coordinates of cells annotated as cancer.
2. Cluster them with DBSCAN [Ester et al. 1996]. A cell is a "core" point if at least `minPts` cancer cells (default 20) lie within radius `eps` (default 35 μm); clusters are chains of density-connected core points. Isolated cancer cells are labelled noise and discarded – this is a feature, because single infiltrating or mis-annotated cells should not define a nest.
3. Draw a concave hull (`concaveman`) around each cluster, producing one polygon per tumor nest. `concavity` controls how tightly the hull follows the cells (lower = tighter).
4. Store the polygons in `seurat_obj@misc$cancer_regions` and annotate every cell with `inside_polygon`.

Because each nest gets its own polygon, stroma between nests is correctly classed as *outside* the tumor.

Parameter selection (`optimize_dbscan_params()`). A grid search over `eps` $\times$ `minPts`, scoring each clustering by the mean silhouette width of the cancer-cell coordinates and reporting number of clusters and fraction of noise. The vignette additionally scores candidate polygon sets by **coverage** (fraction of cancer cells that fall inside a polygon) and **purity** (fraction of cells inside polygons that are cancer), combined into an F_β score ($\beta = 0.5$, weighting purity more heavily).

Signed boundary distance. Using `calculate_boundary_distances()` or `compute_distance_from_boundary()` for each cell the Euclidean distance to the nearest polygon edge is computed and given a negative sign if the cell lies inside any polygon. This single number is the coordinate system for everything that follows.

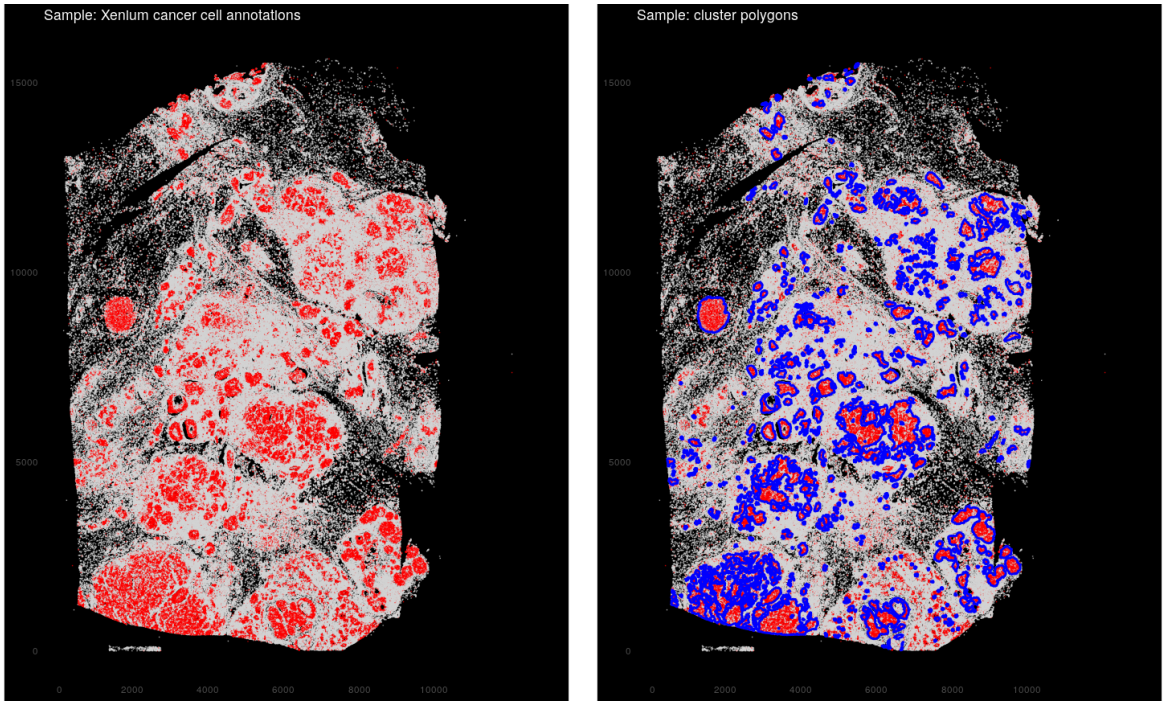


Figure 1: Example: cancer-cell annotations (left) and DBSCAN-derived nest polygons (right).

How to read it, and what to watch for

- **Silhouette score is a geometric, not biological, criterion.** It rewards compact, well-separated blobs. It does not know whether the resulting polygons follow the histological tumor edge. Always inspect the polygons on the tissue and favour the coverage/purity view, with purity weighted more heavily, because impure polygons contaminate the "inside" reference.
- **The boundary inherits the quality of the `is_cancer` label.** On targeted panels, label transfer (e.g. SingleR against an atlas) can misclassify epithelial or proliferating cells. Errors at this step propagate everywhere.
- **The boundary is a 2D cut of a 3D object.** A cell 100 μm "outside" a nest in this section may be 10 μm from tumor just above or below the plane. Distance estimates are therefore upper bounds on true 3D distance.
- **Report the parameters.** `eps`, `minPts` and `concavity` should be stated in any figure legend; a sensitivity check over a few nearby settings is cheap and strongly recommended.

3. Cell-type distributions inside vs. outside the tumor

Why do it

Before modelling gradients, one should describe the two compartments. This answers the first-order questions:

- **Is the ROI what we think it is?** The inside compartment should be dominated by cancer cells; a large non-malignant fraction inside means the polygons are too loose (or the tumor is genuinely highly infiltrated).
- **What is the immune phenotype?** The relative abundance of T cells, B cells, myeloid cells and fibroblasts inside versus outside is the most direct, interpretable readout of inflamed vs. excluded vs. desert biology. A tumor with abundant $CD8^+$ T cells outside but very few inside is the textbook picture of exclusion.
- **What populations are available for downstream analyses?** PCF and TLS analyses require adequate counts of each cell type; this step tells you which comparisons are powered.

What the package does

All three functions operate on a per-cell table of ROI membership and cell type:

- `plot_roi_celltype_dist()` – counts (or proportions) of each cell type among cells **inside** the ROI only, as a ranked horizontal bar chart.
- `plot_roi_inout_celltype_stack()` – 100%-stacked (or count-stacked) bars for inside vs. outside, showing the full composition of each compartment.
- `plot_roi_inout_celltype_dodge()` – side-by-side bars per cell type, which makes per-type inside/outside differences easier to compare than in a stacked chart.

These are descriptive tabulations; no statistical model is applied.

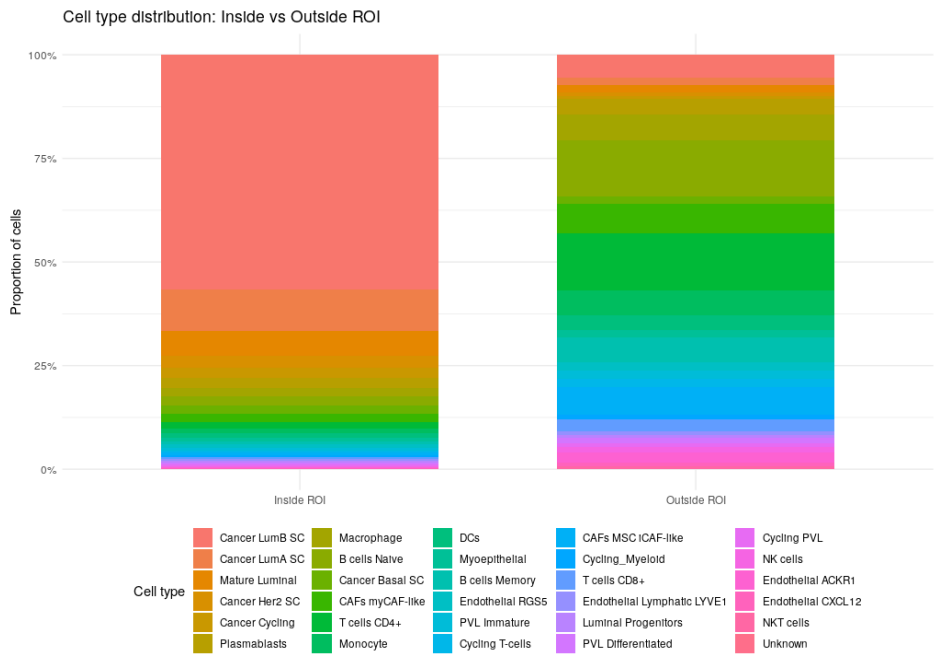


Figure 2: Example: cell-type composition inside vs. outside the ROI.

How to read it, and what to watch for

- **Proportions and counts answer different questions.** Proportions are compositional: if cancer cells dominate the inside, every other type looks rare inside even if its absolute density is unchanged. For infiltration questions, consider densities (cells per mm^2 of each compartment) as well.
- **"Outside" is heterogeneous.** It mixes immediately adjacent peritumoral stroma with distant normal tissue, fat or lymph-node-like areas. The distance metrics in Section 4 exist precisely to resolve this.
- **Segmentation spill-over** at the interface can produce hybrid profiles (e.g. stromal cells with epithelial transcripts), which inflates apparent "cancer cells outside" and "stromal cells inside" near the edge.

4. The three distance metrics and the critical distance

Why do it

The central scientific question of the package is: **how far does the tumor's influence extend into the surrounding tissue, and is the transition sharp or gradual?**

Biologically, this is the width of the peritumoral niche – the zone in which cancer-derived signals (cytokines, growth factors, hypoxia, matrix remodelling, metabolic competition) shape neighbouring cells. Its width is not known *a priori* and varies between tumors. A sharp, narrow transition suggests a well-demarcated, pushing margin; a broad gradient suggests an infiltrative front or long-range paracrine influence. Knowing this distance also gives a principled, data-driven definition of "peritumoral" for downstream analyses, instead of an arbitrary 50 or 100 μm cut-off.

The approach is to ask, at each candidate distance d, *how different are the cells just outside the tumor from the cells inside it?* and to watch how that separation changes as d grows. Three complementary measures of "different" are used, because tumors and stroma can differ in who is present (composition), in what those cells are doing (expression), or both.

What the package does: `calculate_distance_profile()`

For each threshold d in a grid (default 10–200 μm in 10 μm steps):

- **Outside group:** all cells with signed distance in $(0, d]$. Note this is a *cumulative* shell from the boundary out to d, not a thin annulus, so each step adds cells to the previous set.
- **Inside group** (choice of `inside_mode`):
  - `"all"` – every cell inside the tumor polygons (a fixed reference).
  - `"distance"` – only inside cells within d of the boundary, giving a symmetric band of width d on each side.
- Thresholds with fewer than 10 cells in either group are skipped.

Three metrics are then computed for the two groups.

Metric 1 – Transcriptomic distance

$$D_{\text{tx}} = 1 - \text{cor}(\bar{x}_{\text{inside}},\ \bar{x}_{\text{outside}})$$

The mean log-normalised expression of every gene is computed for each group ("pseudobulk" profiles) and the Pearson correlation between the two gene vectors is taken. $D_{\text{tx}} = 0$ means the two compartments have the same overall expression profile; larger values mean greater divergence.

Why: it is a single, global, annotation-free summary of how different the two sides look transcriptionally. It does not depend on cell-type labels, so it is robust to annotation errors.

Caveats: Pearson correlation of mean profiles is dominated by the most highly expressed genes and is typically compressed close to 1, so absolute values are small and differences subtle – interpret the *shape* of the curve, not its magnitude. Because it is a pseudobulk measure, it mixes changes in cell-type composition with changes in cell state.

Metric 2 – Cell-type composition difference

$$D_{\text{comp}} = \sum_k \left(p_k^{\text{inside}} - p_k^{\text{outside}}\right)^2$$

where p_k is the proportion of cell type k in each group (the squared Euclidean distance between composition vectors). Each term is retained as that cell type's **contribution**, and the inside and outside proportions are returned at every distance.

Why: it asks purely "who is there?", and the per-type decomposition tells you *which* populations drive the separation at each distance. The companion plots (`plot_celltype_heatmap()`, `plot_celltype_proportions()`) show, for example, whether fibroblasts dominate the first 30 μm and lymphocytes appear further out – a direct readout of stromal layering and immune exclusion.

Caveats: squared differences are dominated by abundant types (cancer cells, fibroblasts). A biologically decisive but rare population (e.g. B cells or dendritic cells) contributes almost nothing to the total; always inspect the per-type proportion trajectories rather than relying on the total alone. The metric is only as good as the cell-type annotation.

Metric 3 – Differential-expression (DE) strength

Seurat's `FindMarkers()` (Wilcoxon rank-sum, `min.pct = 0.1`, no fold-change threshold) is run for inside vs. outside cells; genes are ranked by $\left|\log_2 \text{FC}\right|$ and the metric is the mean $\left|\log_2 \text{FC}\right|$ of the top 100 genes.

Why: where transcriptomic distance is a global summary, DE strength focuses on the genes that change the most, so it is more sensitive to strong but gene-restricted programmes (e.g. a handful of chemokines or matrix genes) that a whole-transcriptome correlation would wash out.

Caveats: genes are selected by effect size without a significance filter, so the metric can be driven by lowly-expressed, noisy genes. On targeted panels (a few hundred genes) "top 100" is a large fraction of the panel, which makes the metric closer to a panel-wide average. The number of cells changes with d, which affects fold-change estimates for sparse genes. Like Metric 1, it conflates composition and state.

Finding the critical distance: `find_critical_distance()`

For a chosen metric, reports (i) the distance at which separation is maximal and (ii) the **plateau distance**, the smallest d at which the metric reaches 90% of its maximum.

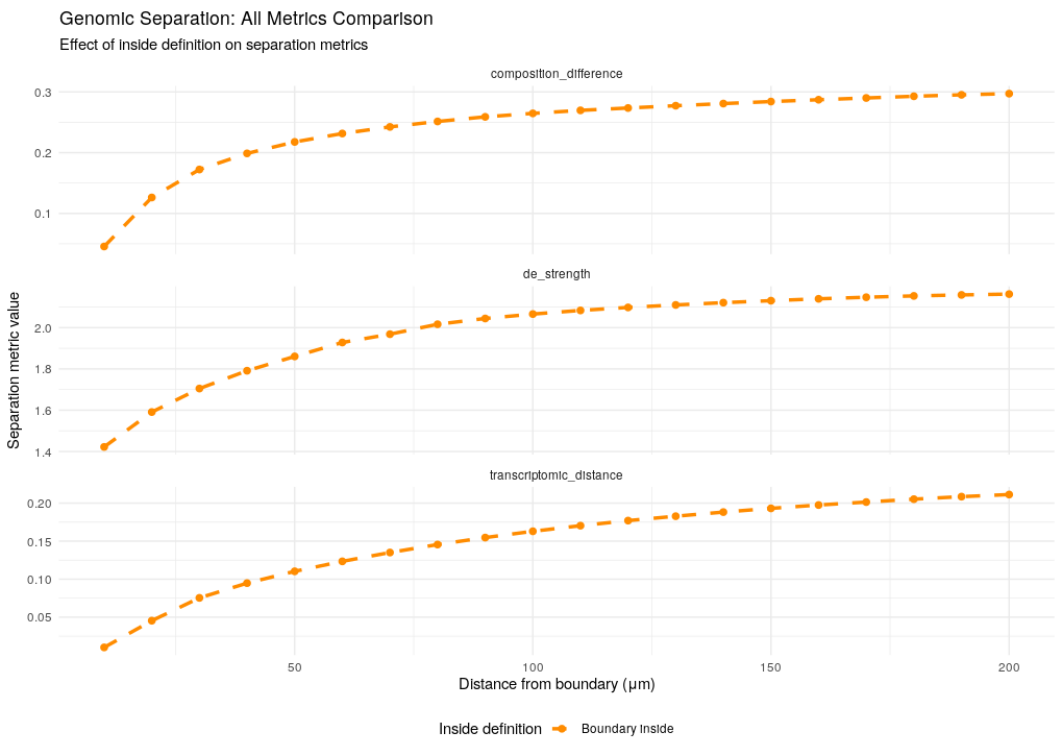


Figure 3: Example: all three metrics as a function of distance from the boundary.

How to read it

- **Shape of the curve.** In `"all"` mode, a typical profile rises as d grows (the outside set incorporates increasingly "stromal" cells that look less like tumor) and then plateaus once additional cells no longer change the outside average. The plateau distance is an operational estimate of the width of the tumor-influenced zone. A steep rise and early plateau implies a sharp boundary; a slow, steady rise implies a broad transition.
- **Comparing inside modes.** `"all"` compares the margin with the *whole* tumor, so it includes core-versus-edge tumor heterogeneity (e.g. hypoxic cores). `"distance"` compares matched bands on either side of the edge, which isolates the interface itself. Divergence between the two curves tells you that the tumor edge differs from the tumor core.
- **Concordance between metrics is informative.** If composition separates strongly but transcriptomic distance does not, the boundary is mainly a change in *who* is present. If transcriptomic distance and DE strength rise while composition is flat, the same cell types are in different *states* near the tumor – often the more interesting biology (e.g. exhausted vs. naive T cells, myofibroblastic vs. inflammatory CAFs).
- **Because the outside set is cumulative**, the curves are smoothed and monotone-leaning; a local effect at, say, 20–40 μm will be diluted at larger d. If you need to localise effects precisely, use annular bins or the gene–distance analysis (Section 6).

5. Pair correlation function (PCF) analysis outside the tumor

Why do it

Composition tells you *who* is present; it does not tell you how cells are *arranged*. Two stromal regions with identical cell-type proportions can be organised very differently: T cells may be uniformly scattered, or tightly clustered with macrophages, or physically segregated from tumor-adjacent fibroblasts. Spatial arrangement is mechanistically meaningful because most immune interactions are contact-dependent or act over a few cell diameters (antigen presentation, co-stimulation, suppressive cytokines). Questions this analysis addresses include:

- Do CD8$^+$ T cells co-localise with dendritic cells (consistent with local priming) or with suppressive macrophages?
- Are T cells segregated from fibroblasts, consistent with a stromal barrier?
- Do B and T cells cluster together (a precursor to TLS)?
- Are particular cell types self-clustered (homotypic aggregation)?

Restricting to cells **outside** the tumor polygons focuses the analysis on the stromal/immune ecosystem around the tumor, and avoids the trivial result that everything appears segregated from the dense cancer-cell mass.

What the package does

1. `build_outside_ppp()` builds a multitype point pattern (`spatstat` `ppp`) from the coordinates of cells outside all polygons, marked by cell type. Types with fewer than `min_cells` (default 10) cells are dropped. The observation window defaults to the bounding box of all cells.
2. `calculate_pcf_outside_summary()` computes the cross-type pair correlation function $g_{ij}(r)$ for every pair of cell types (and, optionally, each type with itself) using `spatstat`'s `pcfcross` with isotropic edge correction, and extracts the value at a chosen distance (default $r = 30$ μm).
3. `plot_pcf_outside_heatmap()` displays these values as a symmetric cell-type $\times$ cell-type heatmap with a diverging scale centred on 1.
4. `plot_pcf_outside_pair_envelopes()` / `plot_pcf_pair_gg()` plot full $g_{ij}(r)$ curves against simulation envelopes (default 10 simulations under complete spatial randomness, bandwidth 10 μm).

Interpretation of $g(r)$. $g_{ij}(r)$ is the density of type-j cells at distance r from a typical type-i cell, divided by the density expected if the two types were placed independently at random.

- $g(r) \approx 1$: no spatial association at distance r.
- $g(r) > 1$: co-localisation / attraction (more neighbours than expected).
- $g(r) < 1$: segregation / avoidance.

Unlike nearest-neighbour or Ripley's K, the PCF is *non-cumulative*, so it shows the specific length scale at which association occurs.

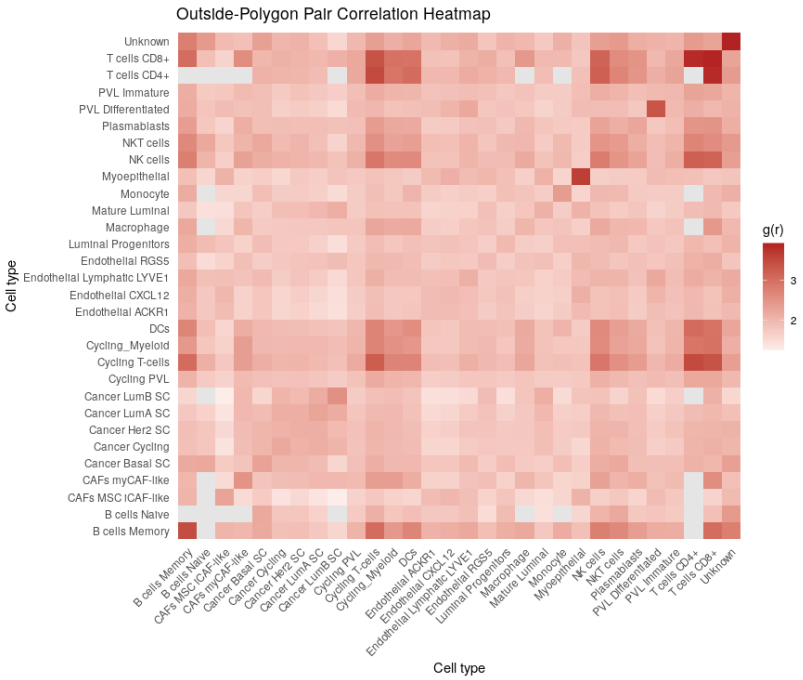


Figure 4: Example: outside-tumor PCF heatmap at r = 30 µm.

How to read it, and what to watch for

- **Choice of window matters.** The default window is the bounding box of the whole section, but by construction there are no "outside" cells inside the tumor polygons (or in empty background). Treating those areas as available space under-estimates the average intensity and pushes $g(r)$ above 1 for *all* pairs. Read the heatmap **relatively** (which pairs are higher than others), or supply a `window_obj` equal to the tissue region minus the tumor polygons.
- **The CSR null asks a narrow question.** Tissue is inherently inhomogeneous, so almost every cell type will appear "clustered" relative to complete spatial randomness. A more biologically pertinent null for cell–cell interaction is *random labelling* (shuffle cell-type labels over the fixed cell positions), which asks whether type j is enriched near type i beyond what tissue density alone would produce.
- **Envelopes with 10 simulations are pointwise and wide.** They correspond to a nominal two-sided significance level of about 0.18 at each r and are not a formal test across r. Use them as a visual guide; increase `nsim` (e.g. 99 or 199) for any claim.
- **Pick r biologically.** 30 μm is roughly two to three cell diameters, a reasonable scale for direct interaction. Inspect the full curves before over-interpreting a single-r heatmap.
- **Rare types are noisy.** PCFs for types near the `min_cells` cut-off have very high variance.

6. Gene–distance analysis

Why do it

The distance metrics in Section 4 tell you *that* and *over what range* the tumor influences its surroundings. The gene–distance analysis tells you *through which genes*. It identifies transcripts whose expression rises or falls continuously with distance from the boundary, in the stroma (peritumoral), in the tumor (intratumoral), or across both.

This is where mechanistic hypotheses come from. Typical findings in this kind of analysis include:

- Genes **decreasing** with distance into the stroma: tumor-induced programmes such as CAF activation and matrix remodelling (e.g. collagens, *POSTN*), angiogenic factors, or immunosuppressive mediators concentrated at the margin.
- Genes **increasing** with distance into the stroma: immune activation or normal-tissue programmes that are suppressed close to the tumor – a molecular signature of exclusion.
- Intratumoral gradients from edge to core: hypoxia, proliferation, or EMT-like programmes at the invasive front.

What the package does

1. `compute_distance_from_boundary()` stores the signed distance (`dist_to_boundary`) for every cell, using the same per-nest polygons as Section 2.
2. `find_distance_associated_genes()`:
   - Selects cells according to `focus`: `"peritumoral"` (distance > 0), `"intratumoral"` (≤ 0, using absolute depth into the tumor), or `"both"` (signed distance across the boundary).
   - Trims the distance range at the smaller of the 95th percentile and `max_distance` (default 200 μm) to avoid sparse, far-field outliers dominating the fit.
   - Keeps genes expressed in at least `min_cells_expressing` (default 10) cells.
   - For each gene, fits an ordinary least-squares model expression $= \beta_0 + \beta_1 \cdot$ distance on log-normalised single-cell expression, recording the slope β_1 (change per μm), its t-statistic, p-value and R^2; and independently computes Spearman's ρ as a rank-based, monotonic, outlier-robust measure. p-values are Benjamini–Hochberg adjusted.
   - For the top significant genes, averages expression within quantile-based distance bins (default 10) to produce trend profiles.
3. `plot_distance_genes()` returns a volcano plot (slope vs. adjusted significance, coloured by direction) and faceted trend plots for the top increasing and decreasing genes.

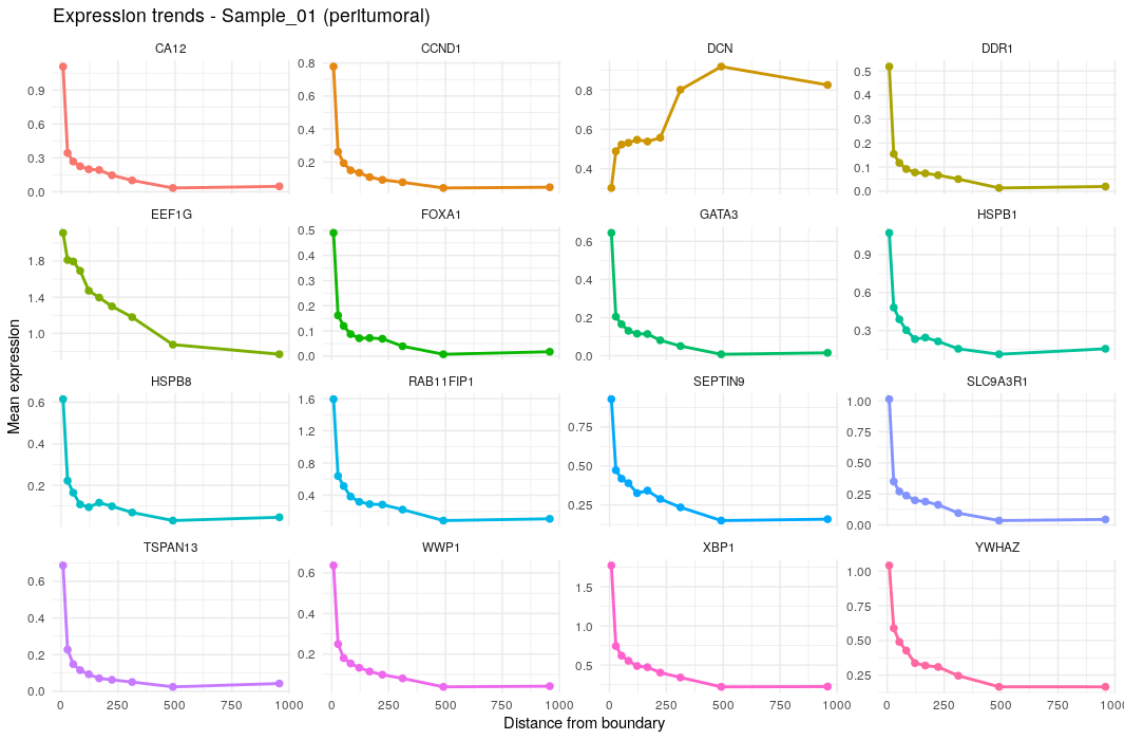


Figure 5: Example: binned expression trends of top distance-associated genes.

How to read it, and what to watch for

- **Prioritise effect size and trend shape over p-values.** With thousands of cells, very small slopes will be "significant". Use β_1, R^2, agreement between the linear and Spearman results, and the visual trends.
- **Composition confounding is the main pitfall.** A gene can appear to decrease with distance simply because the cell types that express it are more abundant near the tumor, without any change in regulation. For example, a fibroblast gene will "decrease with distance" if fibroblasts are concentrated at the margin. To claim a change in cell *state*, repeat the analysis within a single cell type (subset the object first) or include cell type as a covariate.
- **Segmentation spill-over creates artefactual gradients.** In imaging-based platforms, transcripts from a cancer cell can be assigned to an adjacent stromal cell. Tumor-specific genes (e.g. *EPCAM*, keratins) will therefore appear in stromal cells and decay with distance. Treat gradients in canonical tumor markers in the peritumoral analysis as likely technical unless shown otherwise.
- **Linear models assume monotonic, linear trends.** A gene that peaks at 30–50 μm and falls on both sides (a "margin" gene) can have a near-zero slope. The binned trends will reveal such shapes; consider non-linear fits (splines/GAMs) if they matter. In `"both"` mode a single line is fitted across the boundary, which presumes the gene behaves similarly inside and outside.
- **Spatial autocorrelation** means neighbouring cells are not independent, further inflating sig-

nificance. Replication across samples is the real test.

7. Tertiary lymphoid structure (TLS) analysis

Why do it

Tertiary lymphoid structures are ectopic lymphoid organs – organised aggregates of B cells, T cells, dendritic cells and follicular dendritic cells, often with high endothelial venules – that form in chronically inflamed tissues, including tumors. They are sites of local antigen presentation, B-cell maturation and T-cell priming within the tumor microenvironment. Their presence, density and maturity are associated with improved survival and with response to immune checkpoint blockade in several cancers [Sautès-Fridman et al. 2019; Helmink et al. 2020; Cabrita et al. 2020; Petitprez et al. 2020].

Spatial data allow us to do something that dissociated assays cannot: identify individual TLS, measure their composition and maturity, and – combined with the boundary distance – ask *where* they sit relative to the tumor (intratumoral vs. peritumoral vs. distant), which itself appears to carry prognostic information.

What the package does

Structural calling (`call_tls()`).

1. Take B and T cells (labels configurable via `b_types`, `t_types`).
2. Cluster them with DBSCAN (`eps` = 50 μm, `minPts` = 20) to find candidate lymphoid aggregates.
3. Assign *every* cell (of any type) within `eps` of a clustered lymphoid cell to that aggregate, so that stromal, myeloid and dendritic cells embedded in the aggregate are counted.
4. Call an aggregate a TLS if it contains at least `min_B` (20) B cells, at least `min_T` (20) T cells, and B cells make up at least `min_frac_B` (5%) of its cells. The B-cell requirement distinguishes TLS from diffuse T-cell infiltrates.
5. Flag a TLS as **"mature-like"** if it contains any mature/activated dendritic cells (default LAMP3$^+$ or IRF7$^+$ DCs), otherwise **"early/aggregate"**.
6. Store per-aggregate counts, fractions, centroid and approximate area; label cells with `TLS_id` and `in_TLS`.

Transcriptional scoring (`add_tls_score()`). Each cell is scored with `Seurat::AddModuleScore()` for a 22-gene TLS signature spanning B-cell (*MS4A1, CD19, CD22, CD79A/B*), T-cell (*CD3D/E, CD4, CD8A/B, TRBC2, IL7R*), follicular/germinal-centre (*CR2, FCER2, CXCL13, CXCR5, PDCD1*) and antigen-presentation (*CD74, CD1D*) genes. Module scoring subtracts the mean expression of expression-matched control genes, so the score reflects enrichment above background.

Validation and visualisation. `summarize_tls_score()` and `plot_tls_score_comparison()` compare scores in called TLS against the rest of the tissue (Wilcoxon test); `plot_tls_map()`, `plot_tls_score()` and `plot_tls_composition()` show where TLS are, how the signature varies over the tissue, and the cellular make-up of each structure.

How to read it, and what to watch for

- **The structural call and the signature are not independent.** The signature is dominated by B- and T-cell marker genes, and TLS are defined as dense clusters of B and T cells, so a higher score inside called TLS is largely guaranteed. Treat this comparison as a consistency check, not validation. More informative checks are: (i) the follicle-specific genes (*CXCL13, CR2, FCER2, CXCR5*) on their own; (ii) whether high-scoring regions that were *not* called reveal missed TLS; and (iii) comparison with H&E or CD20/CD21 immunostaining.
- **"Mature-like" is a lenient flag.** A single LAMP3$^+$ DC is enough to set it. Histological maturity is usually defined by germinal-centre features – CD21$^+$/CD23$^+$ follicular dendritic cell networks and BCL6$^+$ B cells [Sautès-Fridman et al. 2019]. Where the panel allows, require a minimum number or fraction of DCs, or use FDC/GC markers for a stricter maturity call.
- **Thresholds are tissue- and panel-dependent.** `eps`, `minPts`, `min_B` and `min_T` set the minimum size of what counts as a TLS. Small, early aggregates may be excluded and adjacent TLS can merge. Report thresholds and check that the calls look right on the tissue map.
- **Area is approximate.** It is a bounding-box area, adequate for ranking sizes but not for densities; a hull-based area is more accurate.
- **Connect it to the boundary.** The most informative TLS results usually come from combining them with the rest of the package: TLS density per mm^2 of stroma, the distribution of TLS distances from the tumor edge, and whether TLS-proximal tumor margins show different composition or gene gradients from TLS-distal margins.

8. Putting it together

Analysis	Core question	Unit of comparison	Key output
ROI selection	Where is the tumor?	Cancer-cell coordinates	Nest polygons; signed distance per cell
In/out cell types	Who is on each side?	Inside vs. outside compartment	Composition bar charts
Distance metrics	How far does the tumor's influence extend?	Inside vs. cumulative outside shell at each d	Separation curves; critical/plateau distance
PCF	How are non-tumor cells arranged?	Cell-type pairs outside tumor	$g_{ij}(r)$ heatmap and curves
Gene–distance	Which genes vary with distance?	Single cells along the distance axis	Ranked genes, volcano, trend plots
TLS	Are there organised immune niches?	Lymphoid aggregates	TLS calls, composition, maturity, score

A sensible reading order for a new sample is: validate the boundary (2, 3); estimate the width of the interface (4); use that width to define the peritumoral zone in which to look for gene gradients (6) and spatial organisation (5); and finally place organised immune structures in that context (7).

Recommendations for robust conclusions:

1. Treat each section as one observation; replicate across patients.
2. Report all spatial parameters (`eps`, `minPts`, `concavity`, distance grid, PCF r and window, TLS thresholds) and run small sensitivity analyses.
3. Distinguish **composition** effects from **state** effects – this is the most common source of over-interpretation in boundary analyses.
4. Watch for segmentation spill-over at the interface.
5. Corroborate key findings orthogonally (H&E, immunostaining, or an independent cohort).

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